



Second Level Oversight (SLO) Version Description Document

Foundation Sustainment Services (FS2)

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Revision History

Version #	Implemented By	Revision Date	Reason
1.0.3	Christi Curtis	03/18/2025	Initial release of this version of the product.

Authorization Notification

Any authorization applies only to the current version of the document.

Role	Name	Company	Job Title
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Approval

The undersigned have read this VDD and approve its contents.

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1 Scope

1.1 Introduction

This document is the *Version Description Document (VDD)* for *Aero SLO Version 1.0.3*

This VDD details the software components to be delivered with Aero SLO Version 1.0.3. This document describes the features included in the delivery of the software build, at entrance to test. It includes the build artifacts for deployment, any known issues with the delivery, and the changes included in the software version.

1.2 System Overview

The Second Level Oversight (SLO) tool is a web-based application that monitors and stores the logs of changes done to Airports, Runways, ILS, Nav aids, ATS, and Waypoints in the Aeronautical Application Suite (AAS). The SLO process allows for a double-blind entry of the data to ensure the accuracy of the data entered by analysts. The tool also allows for the export of these changes in the form of an excel spread sheet.

1.3 Document Overview

The Aero SLO VDD provides the list of software delivered for Version 1.0.3 of the product and the capabilities of that software. This document details a single release of the product and will be updated only to correct information specific to version 1.0.3 of the product. Follow-on version of the product will be documented in a separate VDD specific to that version of the product.

The document is organized as follows:

- Section 1 – Scope
- Section 2 – Version Description



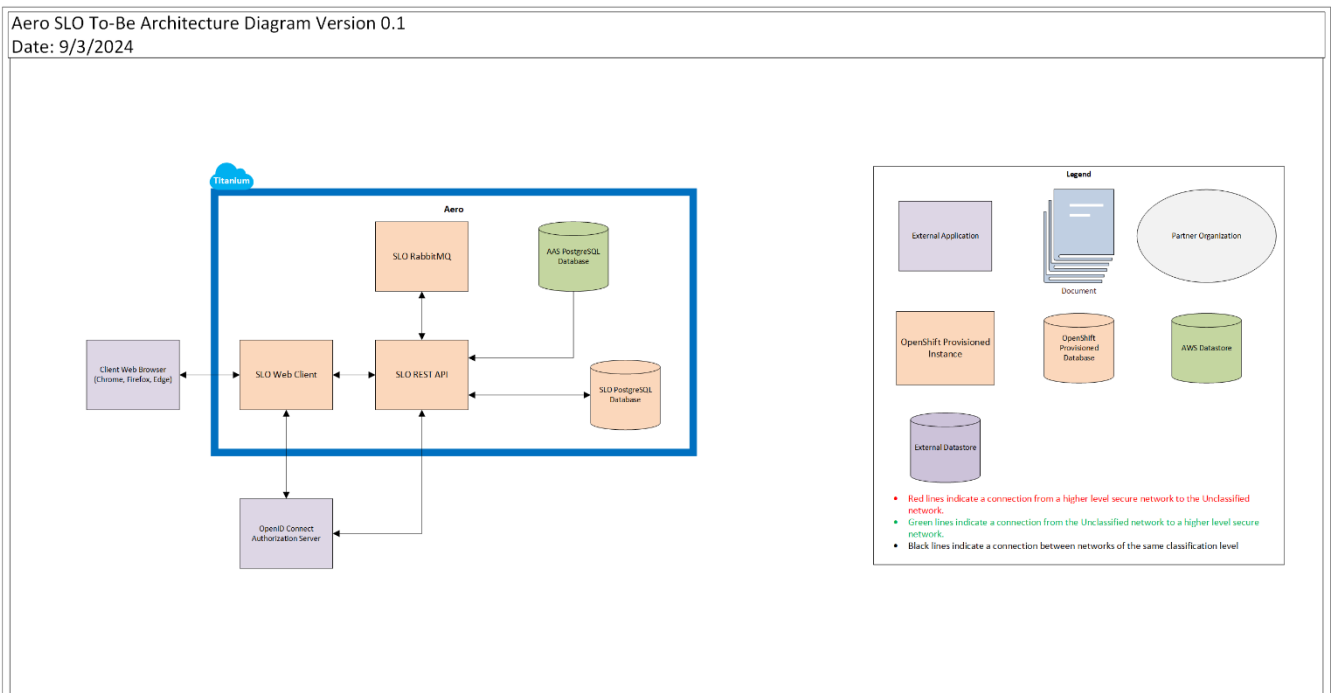
2 Version Description

This release represents an update to the existing SLO application, providing the same basic capabilities that exist in the current release. Code updates have been implemented for ongoing development of features, including the review completion date being added to .csv reports, and the addition of related airport fields information to Navaid reviews. In addition to code updates, several dependency packages were updated to support the SLO database migrations, to address security vulnerabilities, and to replace versions that became disapproved.

2.1 Deployment Environment

The SLO infrastructure will be deployed to the NGA Kubernetes Environment utilizing the OpenShift platform. Relational databases will be deployed to AWS Relational Database Service (RDS) instances. The SLO application components (server, messaging, and web client) will be deployed to OpenShift pods using containers defined by the application team. The following diagram illustrates external connections to services and applications leveraged by the SLO infrastructure, including its connection to aps database.

Architecture Diagram





2.2 Software Functionality and Capabilities

The Aeronautical Application Suite (AAS) provides a single, centralized repository for all National Geospatial-Intelligence Agency (NGA) digital aeronautical data holdings. This system provides analysts with the ability to access and retrieve any stored aeronautical data, and it supports the generation of graphic products. The Second Level Oversight (SLO) process allows for a double-blind entry of the data in AAS to ensure the accuracy of the data entered by the analyst.

2.3 Inventory of Software Components

This section documents the software configuration for SLO.

2.3.1 Software Deliverables

SLO will deliver custom software packages and build artifacts containing the application components, including but not limited to Python files and JavaScript, HTML, and CSS scripts. These packages will include code library dependencies along with custom code to ensure that the application has all the software needed to execute.

2.3.2 Software List

The Aero SLO Version 1.0.3 Software Lists for custom code, COTS and Open-Source code can be found in Table 2.3.2.

Table 2.3.2 - Software List

Software	Version	SWI D	Vendor	Control #	Notes
Second Level Oversight (SLO)	1.0.3	31317	Aeronautical Navigation Office (SFA)	202501300085H	
SLO API					
alembic	1.11.1	16585	Micahel Bayer	20230627051	
amqp (py-amqp)	5.2.0	7005	Python Software Foundation	20240930068	
aniso8601	8.0.0	17379	Python Software Foundation	20210304016	
attrs	21.4.0	18494	Hynek Schlawak	20220919005	
celery	5.4.0	6980	Ask Solem	20240521139	
cfi	1.17.1	15863	Armin Rigo	20241202094	
chardet	4.0.0	7565	Dan Blanchard	20210118085	
click	8.1.3	7573	David Lord	20220922044	
click-didyoumean	0.3.0	27504	Tim Furrer	20230123041	
click-plugins	1.1.1	16725	Kevin Wurster	20200127002	
click-repel	0.2.0	27357	Markus Unterwaditzer	20230123042	
coverage	7.0.1	7609	Ned Batchelder	20221229004	
coverage-badge	1.1.0	19584	Danilo Bargaen	20220803001	
ecs-logging	2.0.0	27563	Elastic NV	20230126023	
et-xmlfile	1.1.0	21401	Charlie Clark	20220307030	
Flask	2.3.3	6810	Armin Ronacher	20231017009	
Flask-Cors	3.0.10	7604	Cory Dolphin	20210412009	
flask-marshmallow	0.14.0	15749	Steven Loria	20221214038	



Flask-Migrate	4.0.0	15911	Miguel Grinberg	20221130029	
flask-restx	1.1.0	20810	Axel Haustant	20230725052	
Flask-Script	2.0.6	19583	Matthias Urlichs	20190726012	
Flask-SQLAlchemy	3.0.2	12583	Armin Ronacher	20221130031	
future	0.18.3	12931	Ed Schofield	20230207043	
gunicorn	20.1.0	5295	Benoit Chesneau	20210517009	
idna	3.3	16052	Kim Davies	20211021035	
importlib-metadata	6.1.0	21335	Jason R Coombs	20230331070	
isort	5.10.1	14649	Timothy Crosley	20220208005	
itsdangerous	2.1.2	7556	The Pallets Project	20221130034	
Jinja2	3.1.2	26785	The Pallets Project	20221111018	
jsonschema	4.4.0	17819	Tom de Grunt	20220523073	
lazy-object-proxy	1.7.1	17381	Ionel Cristian Maries	20220328021	
Mako	1.1.3	16551	Michael Bayer	20210304022	
MarkupSafe	2.1.3	7566	Armin Ronacher	20230724042	
marshmallow	3.14.1	16014	Steven Loria	20220214037	
marshmallow-enum	1.5.1	23835	justanr	20210920006	
marshmallow-sqlalchemy	0.28.2	19591	Steven Loria	20230323084	
mccabe	0.6.1	14654	Ian Cordasc	20170526030	
oauthlib	3.1.0	7784	Ib Lundgren	20191212005	
openpyxl	3.0.10	16698	Eric Gazoni	20220725015	
pipdeptree	1.0.0	16828	Vineet Naik	20210322016	
prompt-toolkit	3.0.36	14746	Jonathan Slenders	20230207047	
psycpg2-binary	2.9.9	32239	Python Package Index (PiP)	20240716043	
pycparser	2.21	15892	Eli Bendersky	20211116084	
PyJWT	2.6.0	14659	Jose Padilla	20221214046	
pylint	2.15.10	13849	Pierre Sassoulas	20230309009	
pylint-flask	0.6	20691	Joe Schafer	20200129023	
pylint-plugin-utils	0.6	24546	Python Code Quality Authority	20220201014	
pyrsistent	0.19.2	19128	Tobias Gustafsson	20221214048	
python-dateutil	2.8.2	6289	Python Software Foundation	20210825079	
pytz	2022.6	7006	Stuart Bishop	20221214049	
requests	2.32.3	6059	Kenneth Reitz	20240617046	
requests-oauthlib	1.3.1	14661	Kenneth Reitz	20220919067	
six	1.16.0	27643	Benjamin Peterson	20230310011	
SQLAlchemy	1.4.46	6982	Michael Bayer	20230123029	
toml	0.10.2	18503	Will Pearson	20211202044	
urllib3	2.2.2	14657	Andrey Petrov	20240617045	
vine	5.1.0	14738	Celery Project	20240731020	
wcwidth	0.2.5	17775	Jeff Quast	20201002017	
Werkzeug	3.0.4	7549	David Lord	20240916047	
zipp	3.12.0	21376	Jason R Coombs	20230208046	
SLO UI					
@elastic/ecs-winston-format	1.3.1	27448	John W Cambra	20230131006	
@emotion/react	11.13.0	24125	Mateusz Burzynski	20240816025	
@emotion/styled	11.13.0	31193	Emma Hamilton	20240816026	
@lagunovsky/redux-react-router	4.3.0	29435	Ivan Lagunovsky	20230814003	



@material-table/core	5.0.2	25172	material-table-core	20230814002	
@mui/base	5.0.0-alpha.76	25190	Olivier Tassinari	20220414018	
@mui/icons-material	5.14.9	24615	Material-UI	20231113037	
@mui/lab	5.0.0-alpha.66	25060	Material-UI	20220125012	
@mui/material	5.16.7	24255	Olivier Tassinari	20240813003	
@mui/styles	5.1.0	24806	Olivier Tassinari	20220317023	
@reduxjs/toolkit	1.9.5	24855	Mark Erikson	20230510051	
axios	1.6.0	7599	Jason Saayman	20231031035	
base-64	1.0.0	18796	Mathias Bynens	20201215051	
Bootstrap (Bootstrap 5 API)	5.2.1-3	24190	Jenkins	20221102022	
core-js	3.33.0	7588	Denis Pushkarev	20231019054	
express	4.18.2	25692	TJ Holowaychuk	20230331037	
history	5.0.0	17133	React Training	20210407013	
isomorphic-fetch	3.0.0	12404	GITHUB/Matt Andrews	20210316010	
jquery	3.6.3	5050	Michal Golebiowski-Owczarek	20230124005	
markdown-to-jsx	7.1.8	18807	Probablyup(Evan Jacobs)	20221221058	
moment	2.30.1	25657	Isken Ivov Chernev	20240216003	
openid-client	4.9.1	22104	Fllip Skokan	20220303081	
prop-types	15.8.1	26615	Facebook	20230615069	
query-string	6.14.1	16601	Sindre Sorhus	20230203036	
react	17.0.2	7572	Dan Abramov	20210604009	
react-bootstrap	2.9.0	20200	Jason Quense	20231025038	
react-datetime	3.2.0	15772	YouCanBookMe	20230203037	
react-dom	17.0.2	31097	Joseph Savona	20210719025	
react-redux	7.2.3	15720	Tim Dorr	20210329012	
react-router	6.10.0	13985	Tim Dorr	20230607058	
react-router-dom	6.10.0	23883	Tim Dorr	20230331038	
redux	4.2.0	14444	DanAbramov	20220525017	
redux-thunk	2.3.0	23701	angular-redux- William Buchwalter	20210816038	
sass	1.57.1	20614	Natalie Weizenbaum	20230203039	
seamless-immutable	7.1.4	22967	Richard Feldman	20210329013	
simple-node-logger	21.8.12	19658	Darryl West	20220601009	
triple-beam	1.4.1	32558	Github/winstonjs	20240814037	
victory	36.6.8	21062	Formidable	20221018021	
winston	3.8.2	10334	Charlie Robbins	20230131004	

2.4 Changes Installed

As was noted in Section 1.3, this is considered to be in initial release of the 1.0.3 Aero SLO product so changes from the previous version are not recorded in this VDD.

Table 2.4 includes defects recorded in testing as well as any other internally discovered defects that were resolved.

- Issue Types:
- Test Problem Report (TPR): DRs written by the NTO during testing
- System Problem Report (SPR): DRs written by problems found by the PMO, customer, or external systems.



Table 2.4 - Changes Installed

PBI/INC #	Key	Issue Type	Summary

2.5 Installation Instructions

The application components will be installed using the NGA CORE services GitLab CI/CD pipeline. No special install instructions will be needed once the pipeline is set up and configured.

2.6 Possible Problems and Known Errors

Table 2.6 Version 1.0.3 Possible Problems and Known Errors

Issue ID	Summary	Description